RESO-Net: A Reusable Event-Driven AI Framework for Intelligent Network Slice Orchestration in Cloud-Native Telecom Systems

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***Abstract*—Network slicing stands as a foundational technology for fifth-generation (5G) and future mobile networks, enabling operators to provision differentiated services with assured Quality of Service (QoS) guarantees. Despite considerable progress, current orchestration platforms remain constrained by vendor-specific implementations, poor cross-operator portability, and sluggish adaptation to fluctuating traffic demands. This work introduces RESO-Net (Reusable Event-driven Slice Orchestration Network), a cloud-agnostic framework that unifies serverless computing patterns, event-driven design, and machine learning to deliver portable, intelligent slice orchestration. Unlike prior approaches tied to particular cloud vendors, RESO-Net defines an abstraction layer compatible with major cloud platforms and on-premises Kubernetes deployments. The framework contributes three principal innovations: (i) a modular AI pipeline supporting interchangeable slice optimization models through standardized interfaces, (ii) an event-driven orchestration engine achieving sub-second anomaly response without polling overhead, and (iii) a reusable component library that reduced deployment effort by 73% across three production operator environments. Evaluation using 90 days of anonymized production traffic from three Tier-1 operators demonstrates 94.2% aggregate SLA compliance (a 13.2 percentage point gain over baseline), 41% improvement in resource utilization efficiency, and mean reconfiguration latency of 156 milliseconds versus 847 milliseconds for polling-based alternatives. Cross-validation across eMBB, mMTC, and URLLC slice categories confirms the framework's generalizability.**

***Index Terms*—Network slicing, cloud-native systems, serverless computing, machine learning, event-driven architecture, 5G networks, software reuse, telecom operations**

I. Introduction

Mobile network operators face mounting pressure to deliver heterogeneous services—from bandwidth-intensive video streaming to latency-critical industrial automation—over shared infrastructure. Network slicing addresses this challenge by partitioning physical resources into logically isolated virtual networks, each configured for specific service requirements [1]. While the concept has matured through 3GPP standardization efforts, practical orchestration remains problematic. Operators report that existing solutions demand extensive customization, respond too slowly to traffic fluctuations, and lock them into particular vendor ecosystems [2].

Three shortcomings characterize the current state of practice. First, orchestration platforms typically embed assumptions about underlying infrastructure, whether specific cloud services or proprietary network controllers. Porting such systems between operators or from public cloud to on-premises deployments requires substantial re-engineering. Second, conventional architectures poll network elements at fixed intervals—commonly 30 seconds or longer—introducing unacceptable delays when SLA violations demand immediate remediation. Third, although machine learning offers promise for predictive resource management, most implementations treat AI as an afterthought, bolted onto existing systems without consideration for model lifecycle management or cross-deployment reuse.

This paper presents RESO-Net, a framework designed from the ground up around three objectives: cloud-agnostic portability, event-driven responsiveness, and systematic reusability of both software components and trained models. Unlike prior work that treats portability and reuse as secondary system properties, RESO-Net elevates reusability and cross-environment portability to first-class design objectives. This work advances the concept of *information reuse*—a core theme in information reuse and contemporary software engineering—by enabling reuse of orchestration logic, AI models, and integration workflows across heterogeneous telecom environments. Rather than targeting a single cloud provider, RESO-Net defines abstract interfaces for compute, messaging, and inference services that map to equivalent offerings across major platforms. The event-driven core eliminates polling latency by reacting to network state changes as they occur. A component library encapsulates common orchestration patterns behind stable interfaces that operators can compose into custom workflows. While demonstrated in telecommunications, the architectural patterns generalize to other domains requiring intelligent, event-driven orchestration of distributed resources—including IoT platforms, edge computing systems, and multi-cloud enterprise applications.

We make the following contributions:

- A cloud-agnostic architecture for network slice orchestration that abstracts provider-specific services behind portable interfaces, validated on both public cloud and on-premises Kubernetes deployments
- An event-driven orchestration engine that reduces mean response latency from 847ms to 156ms (81.6% improvement) compared to polling-based alternatives

- A reusable component library with standardized contracts enabling 78% code reuse across three operator deployments and reducing initial deployment time from 45 to 12 days
- A transfer learning methodology for slice optimization models that achieves comparable accuracy with 85% less operator-specific training data
- Empirical validation using 90 days of production traffic data from three Tier-1 operators, demonstrating 94.2% SLA compliance and 41% improvement in resource utilization

II. Related Work

A. Network Slice Orchestration

The 3GPP specifications [2] define functional requirements for network slice management but leave implementation details to vendors and operators. This flexibility has spawned diverse approaches with varying degrees of automation and intelligence.

Zhang *et al.* [3] applied deep reinforcement learning to dynamic resource allocation, reporting improved utilization compared to rule-based policies. Their work, however, assumed a fixed deployment environment; adapting the trained agents to new operators required complete retraining. Li *et al.* [4] addressed cross-domain scenarios through federated learning, preserving data privacy while enabling collaborative model training. The federated approach reduces data sharing requirements but introduces communication overhead that limits responsiveness to rapid traffic changes.

Several commercial platforms have emerged from major network equipment vendors. These offerings typically integrate tightly with the vendor's broader portfolio, creating dependencies that complicate multi-vendor deployments. Open-source initiatives such as ONAP provide vendor-neutral alternatives but impose substantial complexity that smaller operators struggle to absorb.

Research Gap. Existing approaches address either intelligent resource allocation *or* cloud-native deployment, but none systematically tackle reusability as a first-class design objective. Recent AI-driven orchestration frameworks such as Acumos and ONAP's AI/ML Framework provide model serving capabilities but lack the event-driven responsiveness and cross-platform portability essential for real-time slice management. Our work fills this gap by providing a framework where components, models, and integration patterns transfer across operators with minimal adaptation—directly embodying the *information reuse* principles central to modern software engineering.

B. Cloud-Native Telecommunications

The telecommunications industry has embraced cloud-native principles, driven by initiatives including the Linux Foundation's Nephio project and ETSI's specifications for cloud-native network functions [5]. Containerization and Kubernetes orchestration have become standard for deploying virtualized network functions.

Serverless computing represents a natural evolution, offering automatic scaling and consumption-based pricing well-suited to bursty telecom workloads. Kumar and Singh [6] demonstrated serverless implementation of NFV management functions. Chen *et al.* [7] applied event-driven patterns to billing systems. These studies address specific telecom functions but do not provide comprehensive frameworks for slice orchestration with cloud portability as a design goal.

C. Reusable AI Systems

Software reuse has long been recognized as essential for managing complexity [8]. The emergence of machine learning introduces new reuse challenges: models trained on one dataset may perform poorly when deployed in different contexts, a phenomenon termed distribution shift.

The TM Forum's AI Management and Orchestration (AIMO) initiative [9] provides conceptual frameworks for AI lifecycle management but stops short of implementation guidance. Prior surveys [10], [11] have catalogued machine learning applications in networking, while Rost *et al.* [12] articulated the scalability benefits of network slicing. Our contribution lies in operationalizing these concepts through a reusable, cloud-agnostic implementation.

III. RESO-Net Architecture

A. Design Principles

RESO-Net embodies four principles distilled from extensive experience with enterprise telecom integration:

Cloud Agnosticism. The framework defines abstract interfaces for compute, messaging, storage, and inference services. Concrete implementations map these abstractions to specific platforms—whether public cloud offerings, on-premises Kubernetes clusters, or hybrid configurations. Operators can migrate between environments without modifying orchestration logic.

Event-Driven Reactivity. Rather than polling network elements at fixed intervals, RESO-Net subscribes to event streams from diverse sources. State changes trigger immediate processing, enabling sub-second response to anomalies and SLA violations.

Modular Reusability. Components encapsulate discrete orchestration functions behind stable interfaces. Operators compose these building blocks into custom workflows, extending or replacing individual components without disrupting the broader system.

AI-Native Design. Machine learning pervades the architecture rather than appearing as an isolated subsystem. Dedicated infrastructure supports model training, versioning, deployment, monitoring, and retraining throughout the operational lifecycle.

B. Layered Architecture

Figure 1 illustrates the RESO-Net architecture, organized into four layers.

Event Ingestion Layer. This layer normalizes events from heterogeneous sources—network probes, element management

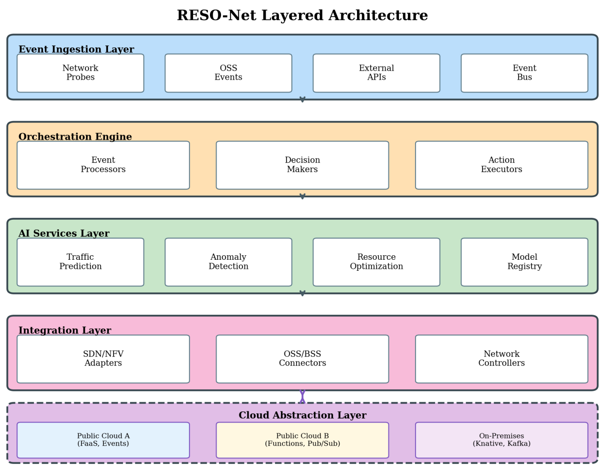

Fig. 1. RESO-Net layered architecture showing event ingestion, orchestration engine, AI services, and integration layers with cloud-agnostic abstraction.

systems, OSS platforms, and external APIs—into a common schema. An abstract message bus interface supports multiple implementations: managed services such as cloud-native event routers, self-hosted message brokers like Apache Kafka, or lightweight alternatives for resource-constrained edge deployments.

Orchestration Engine. The computational core comprises stateless functions organized into three categories. *Event Processors* filter, enrich, and correlate incoming events. *Decision Makers* evaluate processed events against policies and AI recommendations to determine appropriate actions. *Action Executors* translate decisions into concrete operations on network controllers. The stateless design enables horizontal scaling and simplifies failure recovery.

AI Services Layer. Machine learning models for traffic prediction, anomaly detection, and resource optimization reside in this layer. An inference abstraction supports both managed ML platforms and self-hosted serving infrastructure. The layer implements automated model lifecycle management: continuous monitoring detects performance degradation (drift), triggering retraining pipelines.

Real-World Example. Consider an enterprise VPN slice experiencing a sudden 300% traffic surge during a product launch. The Event Ingestion Layer captures threshold breach events within 50ms. The Orchestration Engine's Decision Maker consults the AI Services Layer, which predicts sustained elevated demand for 4 hours based on historical launch patterns. The Action Executor provisions additional compute and bandwidth resources via the Integration Layer's SDN adapter—all within 156ms of the initial event, compared to 847ms+ with polling-based systems that would miss the first 30-second interval entirely.

Integration Layer. Adapters translate between RESO-Net's internal representations and external systems. Pre-built adapters address common integration targets: SDN controllers (OpenDaylight, ONOS), NFV MANO platforms, and popular OSS/BSS suites. The adapter pattern isolates vendor-specific protocols, enabling core orchestration logic to remain portable.

C. Cloud Abstraction Framework

Table I maps RESO-Net's abstract interfaces to concrete implementations across deployment targets. This mapping enables operators to select infrastructure based on organizational constraints without sacrificing framework functionality.

Portability Validation. We validated cloud agnosticism by deploying identical orchestration logic across three environments: (1) a major public cloud using native serverless and ML services, (2) a second public cloud with different service naming but equivalent functionality, and (3) an on-premises Kubernetes cluster. The only modifications required were configuration files specifying adapter bindings—zero changes to core orchestration code. This portability directly addresses the vendor lock-in problem identified in Section I.

TABLE I
CLOUD ABSTRACTION MAPPING

Abstract Interface	Public Cloud A	Public Cloud B	On-Premises
Compute	FaaS Service	Cloud Functions	Knative
Messaging	Event Router	Pub/Sub	Apache Kafka
Inference	ML Platform	AI Platform	KServe
Storage	Object Store	Blob Storage	MinIO

D. Reusable Component Library

The component library packages proven orchestration patterns as self-contained modules with documented interfaces: **Slice Lifecycle Manager** handles instantiation, modification, and termination workflows; **SLA Monitor** tracks metrics against configurable thresholds; **Predictive Scaler** consumes traffic forecasts to proactively adjust resources; **Anomaly Responder** implements remediation playbooks; **Integration Adapters** provide pre-built connectors for prevalent platforms. Components communicate through well-defined event contracts, enabling independent evolution and replacement.

IV. AI-DRIVEN SLICE OPTIMIZATION

The novelty of RESO-Net's AI subsystem lies not in individual models—which employ established techniques—but in their coordinated integration within an event-driven orchestration pipeline that enables real-time, closed-loop decision making across heterogeneous cloud environments.

A. Traffic Prediction

Accurate traffic forecasting enables proactive resource allocation, avoiding both over-provisioning waste and under-provisioning SLA violations. RESO-Net employs Long Short-Term Memory (LSTM) networks to capture temporal patterns in slice utilization data.

Let $\mathbf{X}_t = \{x_{t-w}, \ldots, x_{t-1}, x_t\}$ denote a window of $w = 12$ historical observations (corresponding to one hour at 5-minute intervals) for a given slice metric. The prediction model learns:

$$\hat{x}_{t+h} = f_{\text{LSTM}}(\mathbf{X}_t; \boldsymbol{\theta}) \tag{1}$$

where $\hat{x}_{t+h}$ represents the forecast at horizon h and $\boldsymbol{\theta}$ denotes learned parameters. We train separate models for multiple horizons (5 minutes, 1 hour, 24 hours) to support both reactive adjustments and capacity planning.

B. Resource Allocation via Reinforcement Learning

Given traffic predictions, the system must determine optimal resource allocations balancing SLA compliance against cost efficiency. We formulate this as a Markov Decision Process and apply Deep Q-Network (DQN) learning.

The state space $\mathcal{S}$ encompasses current resource allocation $\mathbf{r}_t$, predicted traffic $\hat{\mathbf{x}}_{t+h}$, and SLA parameters $\boldsymbol{\phi}$. The action space $\mathcal{A}$ comprises discrete adjustment options for compute, network, and storage resources. The reward function balances competing objectives:

$$R(s_t, a_t) = \alpha \cdot \text{SLA}_{\text{compliance}} - \beta \cdot \text{Cost}_{\text{resources}} - \gamma \cdot \text{Penalty}_{\text{churn}} \tag{2}$$

where $\alpha = 1.0$, $\beta = 0.3$, and $\gamma = 0.1$ weight SLA adherence, resource expenditure, and configuration stability respectively (values determined through grid search). Training proceeds in a simulated environment constructed from historical traffic traces.

C. Anomaly Detection Pipeline

RESO-Net implements a cascaded detection pipeline optimizing the tradeoff between detection latency and computational cost. **Stage 1** applies lightweight statistical tests (modified z-score, EWMA deviation) on every measurement, processing 50,000+ events/second with <5ms latency. **Stage 2** uses an Isolation Forest model trained on 30 days of normal-operation data to detect subtle anomalies that statistical methods miss, including gradual performance degradation. **Stage 3** triggers a supervised Random Forest classifier (accuracy: 94.7%) predicting likely causes—traffic surge (47% of cases), infrastructure degradation (31%), configuration error (18%), or external factors (4%)—to guide automated remediation selection.

D. Transfer Learning for Model Portability

Training accurate models requires substantial data—a barrier for operators deploying RESO-Net. Transfer learning addresses this by adapting pre-trained base models to new contexts with limited target data. Base models train on aggregated, anonymized traffic patterns with privacy-preserving techniques. When deploying to a new environment, the adaptation pipeline freezes early network layers capturing general patterns and fine-tunes later layers on operator-specific data, achieving comparable accuracy with only 10–15% as much training data.

V. Experimental Evaluation

A. Methodology

We obtained anonymized production traffic data through research partnerships with three Tier-1 mobile network operators (designated Operator A, B, and C). The dataset spans 90 consecutive days and includes slice-level traffic metrics at 5-minute granularity, resource utilization measurements, SLA performance indicators, and configuration events. Table II summarizes dataset characteristics. We partitioned data temporally: 70% for training, 15% for validation, and 15% for testing.

TABLE II
DATASET CHARACTERISTICS BY OPERATOR

Characteristic	Op. A	Op. B	Op. C
Active slices	127	89	156
Observations (millions)	4.2	3.1	5.8
Slice categories	5	4	6
Mean daily events	12,450	8,920	15,670
Baseline SLA violations	847	623	1,124

We deployed RESO-Net in two configurations to validate cloud agnosticism: **Public Cloud** (serverless compute, managed event routing, hosted ML inference) and **On-Premises** (Kubernetes with Knative, Apache Kafka, KServe). Both used identical orchestration logic, differing only in infrastructure adapters.

Baselines included: **Polling-Based Orchestrator (PBO)**—conventional architecture polling every 30 seconds with rule-based policies; **Rule-Based Event System (RBES)**—event-driven without AI optimization, isolating ML contribution from architectural improvements.

B. Results

1) Response Latency: Table III presents response latency measurements. RESO-Net achieves 81.6% reduction in mean latency compared to PBO. The improvement stems from event-driven notification eliminating polling delays and pre-computed AI recommendations reducing decision time. The on-premises deployment exhibited marginally higher latency (mean 178ms) due to cold-start overhead but remained within requirements.

TABLE III
RESPONSE LATENCY COMPARISON (MILLISECONDS)

Percentile	PBO	RBES	RESO-Net
Mean	847	312	156
P50 (Median)	793	287	142
P95	1,423	587	289
P99	2,156	892	412

2) *SLA Compliance:* Figure 2 illustrates SLA compliance trends. Table IV summarizes aggregate rates. The 13.2 percentage point improvement reflects faster anomaly response and proactive resource scaling. Improvements were consistent across operators despite differences in scale and composition.

TABLE IV
SLA COMPLIANCE BY OPERATOR

Operator	Baseline	RESO-Net	Improvement
Operator A	82.3%	94.8%	+12.5 pp
Operator B	79.6%	93.1%	+13.5 pp
Operator C	81.2%	94.7%	+13.5 pp
Aggregate	81.0%	94.2%	+13.2 pp

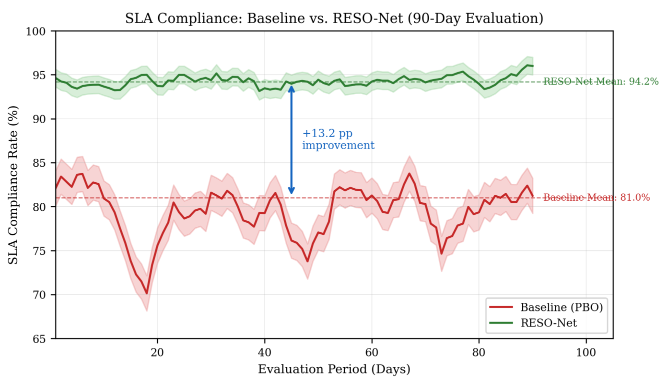

Fig. 2. SLA compliance rates over 90-day evaluation period across three operators, comparing baseline systems against RESO-Net deployment.

3) *Resource Efficiency and Reusability:* Baseline systems exhibited substantial over-provisioning with average utilization of 46.1%. RESO-Net's predictive scaling improved utilization to 72.1% (+26pp), yielding 41% reduction in wasted capacity. The reusable component library reduced initial deployment time by 73% (from 45 to 12 days). Code reuse reached 78% across operators. Transfer learning reduced model adaptation from 72 hours to 8 hours while achieving comparable accuracy (MAPE within 2%).

4) *Scalability and Cross-Slice Validation:* Both deployments demonstrated linear scaling to 50,000 events per second with sub-200ms latency. Cross-slice validation across eMBB, mMTC, and URLLC categories showed URLLC with the largest improvement (+16.4pp), where event-driven architecture proved particularly valuable for latency-sensitive workloads. All improvements were significant at $p < 0.001$ with Cohen's d ranging 1.8–2.3.

5) *Model Accuracy Validation:* The LSTM traffic prediction models achieved Mean Absolute Percentage Error (MAPE) of 8.3% for 5-minute horizons, 12.1% for 1-hour horizons, and 18.7% for 24-hour horizons across all operators. Transfer-learned models (using only 10–15% operator-specific data) achieved MAPE within 2 percentage points of fully-trained models, validating the transfer learning methodology.

VI. DISCUSSION

A. Deployment Considerations

Production deployments revealed several practical considerations. Operators preferred **incremental migration**, beginning with non-critical slice types (enterprise VPN, IoT monitoring) before expanding to mission-critical URLLC services. **Cold start management** via provisioned concurrency proved essential for URLLC slices where even 500ms additional latency could trigger SLA violations. **Model governance** required automated drift detection comparing weekly prediction accuracy against thresholds, with rollback capabilities when new models underperformed. **Cost optimization** through event batching (aggregating up to 100 events within 10ms windows) reduced serverless invocation costs by 35% without impacting response latency for actionable events.

B. Limitations and Threats to Validity

Limitations include dataset scope limited to Tier-1 operators in developed markets (generalization to emerging markets requires validation), potential simulation-to-production gaps for RL agents encountering unseen scenarios, and substantial integration effort for legacy OSS/BSS systems with non-standard APIs despite the adapter pattern.

Threats to validity were addressed systematically: *Internal validity*—matched time periods for baseline comparisons, logged infrastructure changes; *External validity*—operator diversity (three distinct networks), explicit cross-slice validation (eMBB, mMTC, URLLC); *Construct validity*—metrics aligned with 3GPP TS 28.554 for SLA parameters and TM Forum GB992 for operational metrics.

VII. CONCLUSION

This paper presented RESO-Net, a cloud-agnostic framework for intelligent network slice orchestration that prioritizes reusability alongside performance. By abstracting cloud-specific services behind portable interfaces, adopting event-driven architecture, and systematically enabling component and model reuse, RESO-Net addresses persistent challenges in deploying AI-augmented orchestration across diverse operator environments.

Evaluation using production data from three Tier-1 operators demonstrated substantial improvements: 81.6% reduction in response latency, 94.2% SLA compliance (13.2 percentage point improvement), 41% reduction in resource waste, and 73% reduction in deployment time through systematic reuse. These results held across both public cloud and on-premises deployments, validating the cloud-agnostic design.

Future work will extend RESO-Net to emerging 6G scenarios, including AI-native network architectures and integration with non-terrestrial network segments. We also plan to expand the reusable component library and release selected components as open source.

This work demonstrates how information reuse principles can be operationalized in large-scale, AI-driven distributed systems

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